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Russo et al.

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- (54) **CHEMICAL COMPOUND USEFUL AS INTERMEDIATE FOR PREPARING A CATECHOL-O-METHYLTRANSFERASE INHIBITOR**
- (71) Applicant: **BIAL—Portela & C^a, S.A.**, S. Mamede do Coronado (PT)
- (72) Inventors: **Domenico Russo**, S. Mamede do Coronado (PT); **Laszlo Erno Kiss**, S. Mamede do Coronado (PT); **Jorge Bruno Reis Wahnnon**, S. Mamede do Coronado (PT); **David Alexander Learmonth**, Alfena (PT); **Tibor Eszenyi**, Tiszaok (HU); **Axel Zimmermann**, Kinkel-Altstadt (DE); **Bjoern Schlummer**, Leverkusen (DE); **Michael Kreis**, Leverkusen (DE); **Klaus Reiter**, Linz (AT)

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- (63) Continuation of application No. 14/365,265, filed as application No. PCT/PT2012/000048 on Dec. 12, 2012, now Pat. No. 9,126,988.
- (60) Provisional application No. 61/718,589, filed on Oct. 25, 2012, provisional application No. 61/593,625, filed on Feb. 1, 2012, provisional application No. 61/570,141, filed on Dec. 13, 2011.

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- (52) **U.S. Cl.**
CPC **C07D 413/04** (2013.01)
- (58) **Field of Classification Search**
None
See application file for complete search history.

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Primary Examiner — Heidi Reese
(74) *Attorney, Agent, or Firm* — McCarter & English, LLP; Steven G. Davis; Wei Song

(57) **ABSTRACT**

There is disclosed a methylated intermediate which may be demethylated to provide an inhibitor of catechol-O-methyltransferase useful in the treatment of Parkinson's disease. Also disclosed are methods of making and using said intermediate.

12 Claims, 6 Drawing Sheets

Fig. 1

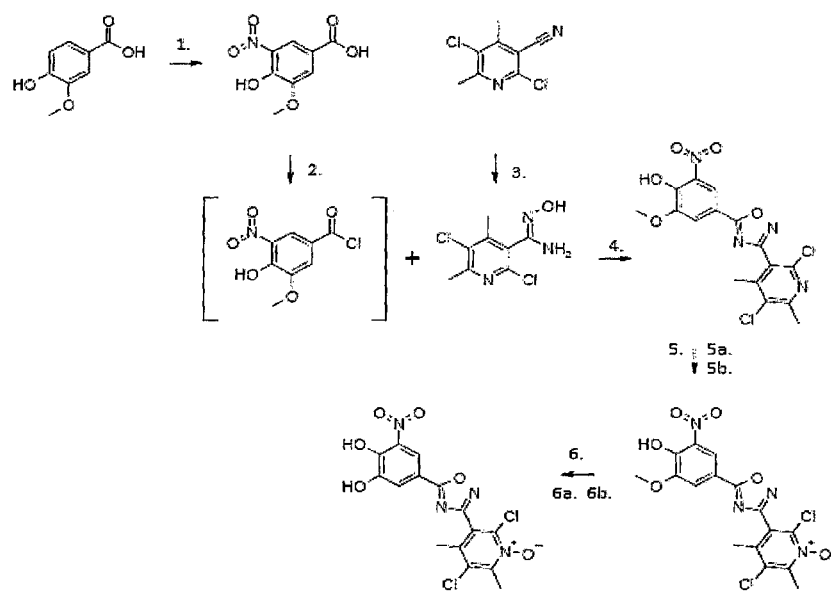


Fig. 2

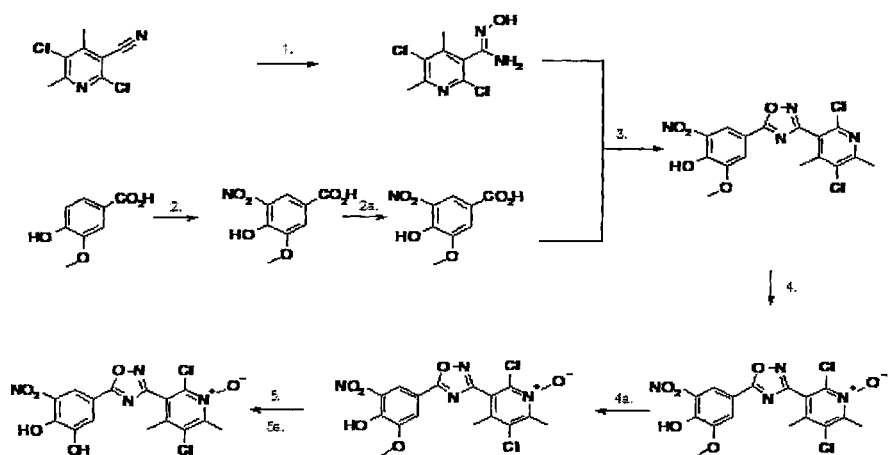


Fig. 3

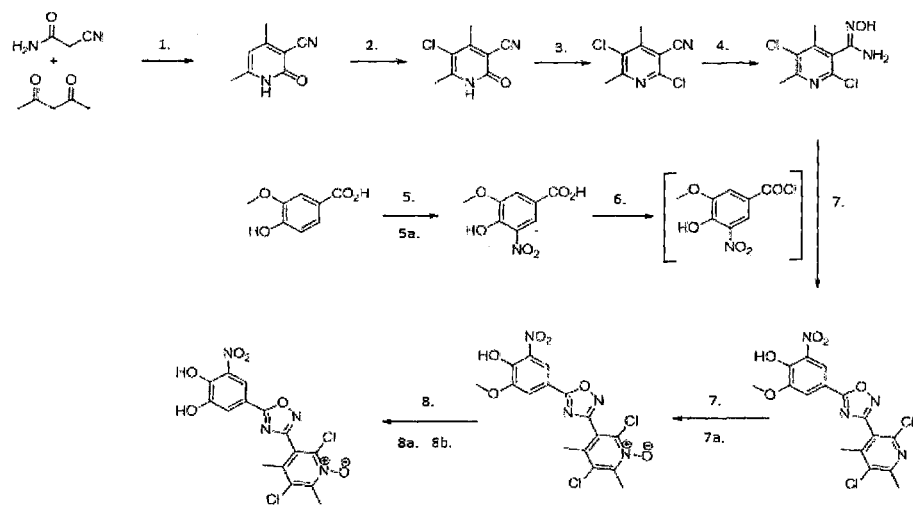


Fig. 4

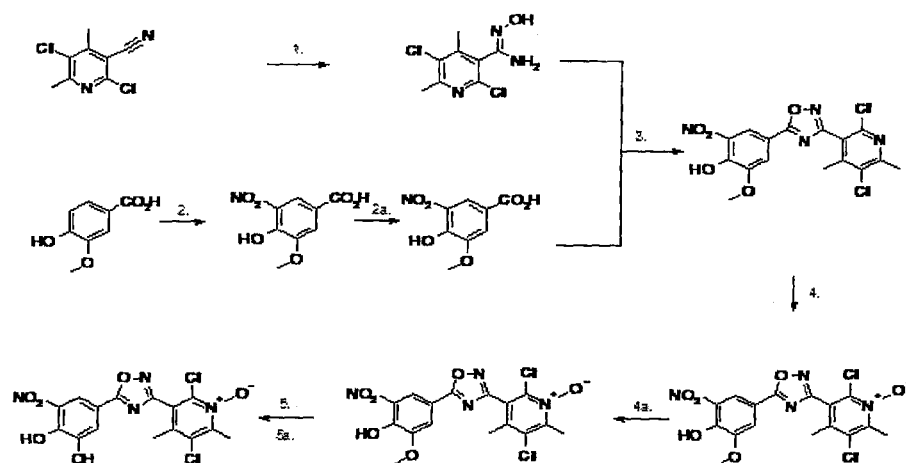


Fig. 5

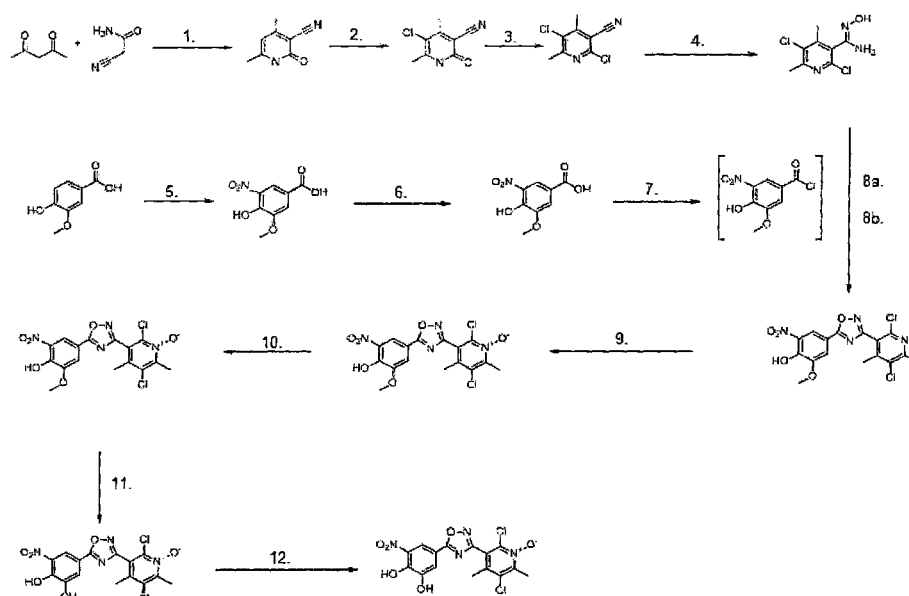
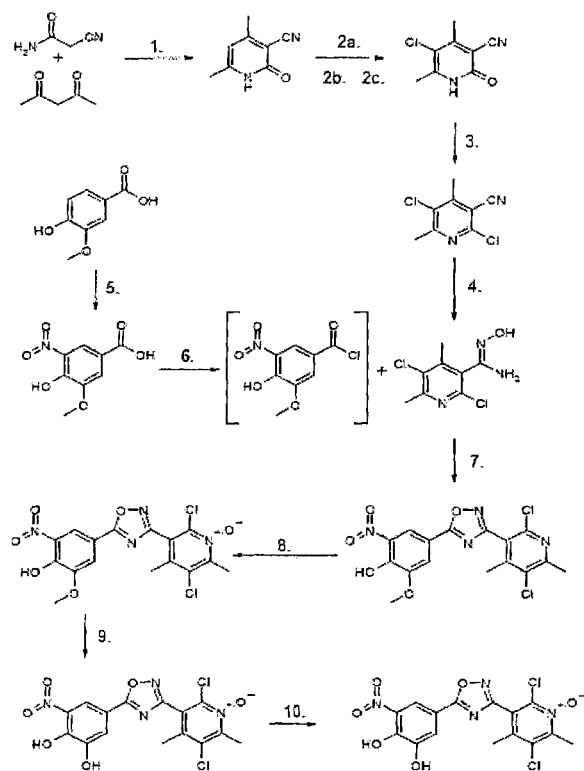


Fig. 6



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CHEMICAL COMPOUND USEFUL AS INTERMEDIATE FOR PREPARING A CATECHOL-O-METHYLTRANSFERASE INHIBITOR

RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 14/365,265, filed Jun. 13, 2014, which is a 35 U.S.C. §371 national stage filing of International Application No. PCT/PT2012/000048, filed Dec. 12, 2012, which claims priority under 35 U.S.C. §365 to Great Britain Patent Application No. 1201758.8, filed on Feb. 1, 2012; Great Britain Patent Application No. 1121413.7, filed on Dec. 13, 2011; U.S. Provisional Application No. 61/718,589, filed on Oct. 25, 2012; U.S. Provisional Application No. 61/593,625, filed on Feb. 1, 2012; and U.S. Provisional Patent Application No. 61/570,141, filed on Dec. 13, 2011. The entire contents of each of the aforementioned applications are incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates to a novel compound and to processes which employ it in the preparation of a catechol-O-methyltransferase inhibitor. In particular this invention relates to 5-[3-(2,5-dichloro-4,6-dimethyl-1-oxy-pyridin-3-yl)-[1,2,4]oxadiazol-5-yl]-2-hydroxy-3-methoxy-1-nitrobenzene which can be used in the process for the preparation of 5-[3-(2,5-dichloro-4,6-dimethyl-1-oxy-pyridin-3-yl)-[1,2,4]oxadiazol-5-yl]-3-nitrobenzene-1,2-diol.

BACKGROUND TO THE INVENTION

A preferred method of treatment of Parkinson's disease is the administration of a combination of levodopa and a peripherally selective aromatic amino acid decarboxylase inhibitor (AADCi) together with a catechol-O-methyltransferase (COMT) inhibitor. The currently employed COMT inhibitors are tolcapone and entacapone. However, some authorities believe that each of these COMT inhibitors have residual problems relating to pharmacokinetic or pharmacodynamic properties, or to clinical efficiency or safety. Hence, not all patients get most benefit from their levodopa/AADCi/COMT inhibitor therapy.

Favoured new COMT inhibitors were disclosed in L. E. Kiss et al, J. Med. Chem., 2010, 53, 3396-3411 (D1), WO 2007/013830 (D2) and WO 2007/117165 (D3) which are believed to have particularly desirable properties so that patients can benefit from enhanced therapy.

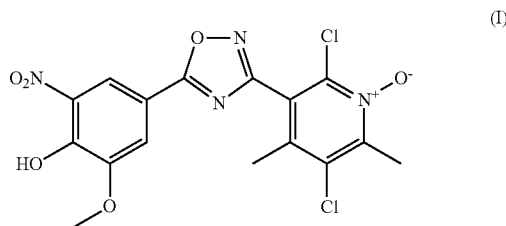
D1, D2 and D3 also disclosed methods of preparing the new COMT inhibitors. Those processes, although effective, would benefit from an increase in yields. Other benefits which would be appropriate include those selected from reduction in number of process steps, reduction in number of unit operations, reduction of cycle-times, increased space yield, increased safety, easier to handle reagents/reactants and/or increase in purity of the COMT inhibitor, especially when manufacture of larger quantities are envisaged. A process has now been discovered that proceeds via a new intermediate which is suitable for manufacture of commercially useful quantities of a particularly apt COMT inhibitor in good yield. Additional benefits occur such as those selected from a reduced number of process steps and number of unit operations, reduced cycle-times, increased space

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yield, increased safety, with easier to handle reagents/reactants, improved impurity profile and/or good purity.

BRIEF DESCRIPTION OF THE INVENTION

The present invention in one aspect provides 5-[3-(2,5-dichloro-4,6-dimethyl-1-oxy-pyridin-3-yl)-[1,2,4]oxadiazol-5-yl]-2-hydroxy-3-methoxy-1-nitrobenzene and salts thereof, that is the compound of the formula (I):

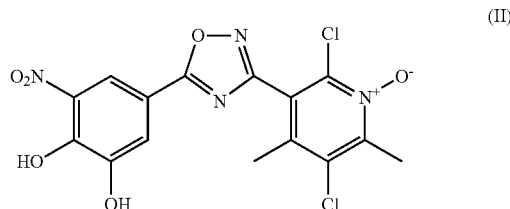


and salts thereof.

Most aptly the compound of formula (I) is unsalted. However, salts of the hydroxy group with metal ions such as the alkali or alkaline earth metals, particularly the sodium and potassium salts are provided as well as those of highly basic organic compounds such as guanidine or the like.

Particularly suitably the compound of formula (I) or its salt is provided in a form suitable for use as a chemical intermediate. This may be, for example, in a form at least 50% pure, in crystalline form, in solid form or in an organic solvent or the like.

The compound of formula (I) is useful as an intermediate in the preparation of 5-[3-(2,5-dichloro-4,6-dimethyl-1-oxy-pyridin-3-yl)-[1,2,4]oxadiazol-5-yl]-3-nitrobenzene-1,2-diol i.e. the compound of formula (II):



The compound of formula (II) may also be referred to as opicapone or 2,5-dichloro-3-(5-(3,4-dihydroxy-5-nitrophenyl)-[1,2,4]oxadiazole-3-yl)-4,6-dimethylpyridine-1-oxide. Opicapone has been found to be more potent than tolcapone in inhibiting liver COMT both at 3 hours and 6 hours post oral administration to rats [ED₅₀ in mg/kg, opicapone 0.87 at 3 hours and 1.12 at 6 hours as compared to tolcapone 1.28 at 3 hours and 2.08 at 6 hours]. Opicapone at a dose of 3 mg/kg was found to be more effective at inhibiting rat liver COMT with nearly complete inhibition occurring 2 to 6 hours post oral administration with only about 90% of enzyme activity recovered after 72 hours while tolcapone provided shorter duration of activity with about 84% recovery after only 9 hours. Both opicapone and tolcapone inhibit human recombinant S-COMT but opicapone has an inhibitory constant of 16 pM being 10 fold lower than that for tolcapone. With respect to the desirable property of avoiding inhibition of COMT in the brain, opicapone following oral administration to the rat was found to be devoid of effect

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whereas tolcapone inhibited about 50% of enzyme activity over a period of 8 hours post administration.

In a further aspect the invention provides a process for the preparation of the compound of the formula (II) as set forth above or a salt thereof which comprises the demethylation of a compound of the formula (I) as set forth above or a salt thereof. In one embodiment, the process does not require compound of the formula (I) to be dried for the subsequent demethylation reaction, i.e. compound of the formula (I) may be wet with toluene.

Typically, the amount of toluene in compound of the formula (I) is in the range 1% to 60% w/w. This is advantageous as it improves the process time and safety. In another embodiment, the process does not require the compound of the formula (II) to be slurried in EtOH. This is advantageous as it improves the process cycle-time.

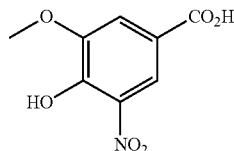
Most aptly the process is adapted to the preparation of a compound of the formula (II) but salts thereof may be prepared, for example an alkali metal or alkaline earth metal salt, preferably the sodium or potassium salt, or a salt of a strongly basic organic compound such as a guanidine.

The O-demethylation reaction may be effected by reaction with a demethylating reagent. A suitable demethylating reagent is a Lewis acid in the presence of appropriate base, for example, aluminium chloride (AlCl_3) and pyridine. The demethylation will generally be performed at a moderately elevated temperature, preferably between 45°C .- 70°C ., more preferably between 55°C .- 65°C .

The compound of the formula (II) prepared by this process can be sufficiently pure for use in a pharmaceutical composition for use in the treatment of Parkinson's disease as hereinbefore indicated. The thus prepared compound of formula (II) may be ball milled or otherwise provided in microparticulate form, for example micronized through jet mills (MC JETMILL®). Thus, in a further aspect of the invention provides a pharmaceutical composition which comprises a compound of formula (II) in microparticulate form for use in the treatment of Parkinson's disease by oral administration.

It is a particular advantage of the present process that the product of the reaction of the compounds of formulas (IV) and (V) obtained after precipitation with ethanol may be employed without the need for isolation of crude compound of formula (III) as the work-up procedure allows the isolation of compound of formula (III) with a purity not less than 95% (HPLC), preferably not less than 96% and ready to use in the next stage of the synthesis. Another advantage of the present process is the optional ability to omit the isolation of any intermediate compounds of the reaction of the compounds of formulas (IV) and (V).

In one embodiment, the acyl chloride may be prepared by the reaction of the compound of formula (VI):



with SOCl_2 in dioxane at 75°C .- 85°C . The use of dioxane in this stage facilitates ease of use in subsequent reactions, for example if it occurs in a solvent system containing dioxane (no need for solvent swap before proceeding with the next step), allows a higher space yield, higher process

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output, requires less reaction vessels, shorter reaction times, improved solubility of reactants (homogeneous reaction solutions mixture instead of a slurry) and avoids the use of DMF (increased purity).

In another embodiment, the reaction of compound of formula (VI) with SOCl_2 is performed in DCM in the presence of a catalytic amount of DMF at 35 - 50°C ., preferably at reflux temperature.

When preparing the compound of formula (II) in a form for use in a pharmaceutical composition, it may be recrystallized from propan-2-ol and formic acid and thereafter ball milled or micronized through spiral jet mills to provide particles of the desired size for good oral bioavailability and/or suitable properties (e.g. suitable particle size) for the preparation of a pharmaceutical composition.

BRIEF DESCRIPTION OF THE FIGURES

Acronyms List

DMF—Dimethylformamide
 SOCl_2 —Thionyl Chloride
 MeOH—Methanol
 THF—Tetrahydrofuran
 DMac—Dimethyl acetamide
 TFAA—Trifluoroacetic acid anhydride
 IPA—Isopropoanol
 HNO_3 —Nitric acid
 DCM—Dichloromethane
 EtOH—Ethanol
 HCl—Hydrochloric acid
 UHP—Urea Hydrogen Peroxide
 AlCl_3 —Aluminium trichloride
 NMP—N-methylpyrrolidone
 POCl_3 —Phosphoryl chloride
 $(\text{CH}_3)_4\text{NCl}$ —Tetramethylammonium chloride

FIG. 1. Process to prepare compound of formula (II) using compound of formula (I) as an intermediate according to one embodiment of the invention.

1. nitric acid 65%, acetic acid, 10 - 20°C ., recrystallization; 2. SOCl_2 , DMF (catalytic), 50°C .; 3. 50% hydroxylamine in water, catalytic amount of 1,10-phenantroline hydrate, MeOH, 75 - 80°C .; 4. THF, DMac, pyridine, 110 - 120°C .; 5. TFAA, DCM, UHP, 10 - 20°C .; 5a. solvent swap from DCM to acetonitrile; 5b. crystallization from toluene/formic acid; 6. aluminium chloride, pyridine, N-methylpyrrolidone; 6a. Ethanol reslurry; 6b. Re-crystallization from IPA/formic acid.

FIG. 2. Process to prepare compound of formula (II) using compound of formula (I) as an intermediate according to one embodiment of the invention.

1. 50% NH_2OH in water, catalytic amount of 1,10-phenantroline hydrate MeOH; 2. 65% HNO_3 , acetic acid, 2a. Re-crystallization from acetic acid; 3. SOCl_2 , DCM, DMF (catalytic), solvent switch from DCM to THF, addition of acid chloride to amidoxime in DMac, addition of pyridine, heat to 110°C .; Quench on aq. HCl and DCM; crystallization from DCM/EtOH; 4. DCM, UHP, TFAA, solvent switch from DCM to toluene/formic acid; crystallization from toluene/formic acid; 4a. Re-crystallization from formic acid/toluene; 5. AlCl_3 , NMP, pyridine, the compound of formula (II) is precipitated and isolated by addition of diluted HCl; 5a. Re-crystallization from IPA/formic acid.

FIG. 3. Process to prepare compound of formula (II) using compound of formula (I) as an intermediate according to one embodiment of the invention.

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1. morpholine in MeOH; 2. SOCl_2 in acetonitrile; 3. POCl_3 , $(\text{CH}_3)_4\text{NCl}$ in DCM; 4. 50% NH_2OH in water, catalytic amount of 1,10-phenantroline hydrate, MeOH; 5. 65% HNO_3 , acetic acid, 5a. Re-crystallization from acetic acid; 6. SOCl_2 , DCM, DMF (catalytic), solvent switch from DCM to THF; 6a. addition of acid chloride to amidoxime in DMAc, addition of pyridine, heat to 110°C .; precipitation of DMAc solution aq. HCl with isolation of crude compound of formula (III); crystallization from DCM/EtOH; 7. DCM, UHP, TFAA, solvent switch from DCM to toluene/formic acid; crystallization from toluene/formic acid; 7a. Re-crystallization from formic acid/toluene; 8. AlCl_3 , NMP, pyridine; the compound of formula (II) is precipitated and isolated by addition of diluted HCl, 8a. reslurry in ethanol 8b. Re-crystallization from IPA/formic acid.

FIG. 4. Process to prepare compound of formula (II) using compound of formula (I) as an intermediate according to one embodiment of the invention.

1. 50% NH_2OH in water, catalytic amount of 1,10-phenantroline hydrate MeOH; 2. 65% HNO_3 , acetic acid, 2a. Re-crystallization from acetic acid; 3. SOCl_2 , dioxane, addition of acid chloride to amidoxime in dioxane, addition of pyridine, heat to 110°C .; Quench on aq. HCl and DCM; crystallization from DCM/EtOH; 4. DCM, UHP, TFAA, solvent switch from DCM to toluene/formic acid; crystallization from toluene/formic acid; 4a. Re-crystallization from formic acid/toluene; 5. AlCl_3 , NMP, pyridine; the compound (II) is precipitated and isolated by addition of diluted HCl; 5a. Re-crystallization from IPA/formic acid.

FIG. 5. Process to prepare compound of formula (II) using compound of formula (I) as an intermediate according to one embodiment of the invention.

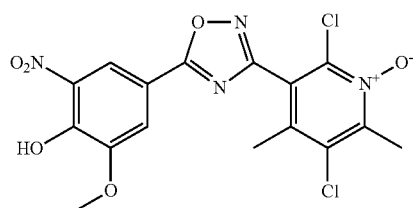
1. morpholine, MeOH, 2. SOCl_2 , 3. POCl_3 ; 4. H_2NOH , 1,10-phenantroline; 5. HNO_3 , acetic acid; 6. recrystallization from acetic acid; 7. SOCl_2 , 1,4-Dioxane; 8a. 1,4-Dioxane, Pyridine; 8b. EtOH; 9. DCM, UHP, TFAA; 10. toluene, formic acid; 11. AlCl_3 , NMP, Pyridine; 12. Formic acid/IPA.

FIG. 6. Process to prepare compound of formula (II) using compound of formula (I) as an intermediate according to one embodiment of the invention.

1. morpholine, MeOH, 22 h, reflux; 2a. SOCl_2 , MeCN; 2b. 2 h, 65°C .; 2c. 2 h, 20°C .; 3. POCl_3 , TMACl, 8 h, 110°C .; 4. $\text{NH}_2\text{OH}/\text{H}_2\text{O}$, 1,10-phenantroline monohydrate, MeOH, 6 h, 75°C .; 5. HNO_3 , HOAc, $10-20^\circ\text{C}$.; 6. SOCl_2 , DCM/DMF, 8 h, 40°C .; 7. DMA/THF, 2 h, $5-10^\circ\text{C}$.; 8. Urea- H_2O_2 , DCM/TFAA, 18 h, 20°C .; 9. AlCl_3 , NMP/Pyridine, 2 h, 60°C .; 10. Recrystallization.

DETAILED DESCRIPTION OF THE INVENTION

The present invention provides the compound of the formula (I):



and salts thereof.

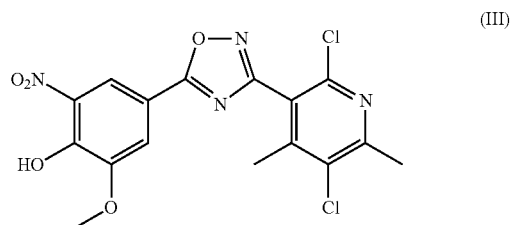
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The use of the compound of formula (I) leads to a particularly effective process for the preparation of the compound of formula (II). By avoiding the deprotection of both phenolic hydroxyl groups of prior art processes, good yields may be achieved when starting from the readily available and relatively less expensive compound vanillic acid.

The compound of formula (I) may be obtained in high purity, for example in crystalline form, which also helps achieve the preparation of the compound of formula (II) in highly pure forms, for example containing only very low amounts of impurities.

Aptly, the compound of formula (I) is crystallised and/or recrystallized from a mixture of organic solvents one of which is an acid, favourably formic acid. A preferred recrystallization solvent for the compound of formula (I) is a mixture of toluene and formic acid. Another preferred recrystallization solvent system for the compound of formula (I) is formic acid/isopropanol (solvent/antisolvent).

The compound of formula (I) or salt thereof may be prepared by the oxidation of the compound of the formula (III):



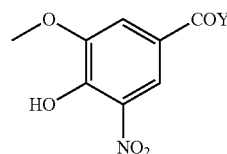
or salt thereof.

Normally and preferably the unsalted compound of formula (I) is prepared from a compound of formula (III) but if a salt is required this may be produced by reaction of the phenolic hydroxyl group with an appropriate base after the formation of the compound of the formula (I).

The oxidation reaction may be performed with any suitable oxidizing agent but preferably a peroxide is employed. Suitably the peroxide may be H_2O_2 which is preferably employed as H_2O_2 -urea addition complex. The oxidation is preferably carried out in the presence of an organic acid anhydride such as trifluoroacetic anhydride.

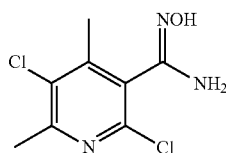
The oxidation generally takes place in a non-hydroxylic organic solvent, preferably in halogenated solvents such as methylene chloride. The oxidation is preferably performed at between 15°C . and 30°C ., more preferably from 20°C .- 25°C .

The compound of formula (III) may be prepared by the reaction of a compound of formula (IV) wherein Y is a halo group, such as chloride, or OR in which R could be hydrogen or a C1-C6 alkyl such as methyl or ethyl:

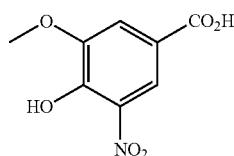


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with a compound of the formula (V):



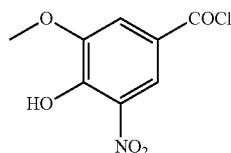
The reaction of the compounds of formula (IV) and (V) can take place in an organic solvent and more generally in a mixture of organic solvents at least one of which will be basic solvent, for example pyridine. A suitable mixed solvent is dimethylacetamide, tetrahydrofuran and pyridine. Alternatively, the solvent organic mixture is a mixture of dioxane and pyridine. The reaction of the compounds of formula (IV) and (V) can also take place in the presence of an organic base such as pyridine or a tertiary amine. When Y is OR and R is hydrogen in the compound of formula (IV), the compound has formula (IX):



When Y is OR and R is hydrogen in the compound of formula (IV), the addition of a coupling reagent such as carbodiimide, phosphonic acid derivatives, carbonyl diimidazole derivatives is required.

When Y is OR and R is C1-C4 alkyl such as methyl, the addition of a Lewis acid such as aluminium trichloride, or a Brönstedt acid such as p-Toluene sulfonic acid catalyst may be required.

When Y is a chloride, compound of formula (VIII), may be preferably used to prepare the compound of formula (III).



The cyclization process will take place at an elevated temperature, for example at 100° C.-120° C. Particularly the process will be performed at 105° C.-115° C.

If desired a further organic liquid such as ethanol may be added at the end of the reaction. Suitably precipitation is not effected by the addition of such a further organic liquid.

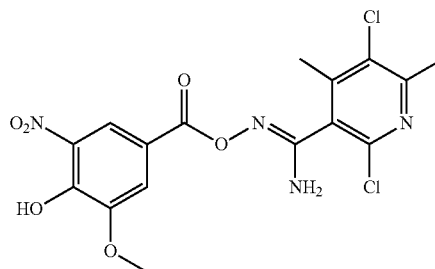
The reaction of the compounds of formulas (IV) and (V) is believed to proceed via the open chain intermediate shown below:

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(V)

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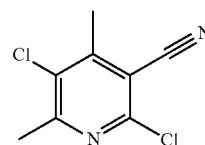
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It is an advantage of the present process that this intermediate need not be isolated but becomes cyclised to the desired compound of formula (III) under the reaction conditions employed. Suitably the reaction is performed at a temperature of between 100-120° C. to give the desired cyclised compound of formula (III).

The use of the compound of formula (VIII) has been found to lead to enhanced yields in comparison with other activated analogues such as those formed from the acid and coupling reagents.

In one embodiment of the present invention the compound of formula (V) is prepared from the compound of the formula (VII)



(VII)

by reaction with hydroxylamine in the presence of 1,10-phenanthroline monohydrate. Preferably the reaction is carried in a mixture of methanol and water at 70-80° C. Conveniently, the presence of 1,10-phenanthroline monohydrate reduces or eliminates the formation of unwanted amide and favours the formation of the wanted amidoxime of formula (V).

The compound of formula (IV) may be prepared from the corresponding carboxylic acid by esterification or formation of an acid chloride. Aptly for preparing an acid chloride this may involve reaction with SOCl₂. Such corresponding acid may be prepared by nitration of vanillic acid. Compounds of Formula (IV) which are esters may also be prepared by nitration of the corresponding ester of vanillic acid, for example by nitration of the methyl ester of vanillic acid. Suitable conditions for such reactions are set forth in the Examples hereinafter.

The compound of formula (II) in crystalline form is particularly apt for use in pharmaceutical compositions for administration orally. In particular, such compositions may be in the form of discrete unit doses such as tablets or capsules.

The pharmaceutical composition which contains the compound of formula (II) in crystalline form, preferably prepared as described herein, will also comprise a carrier therefor. Suitable carriers include those described in D1, D2 or D3, referred to hereinbefore. The pharmaceutical composition which contains the compound of formula (II) in crystalline form may additionally comprise levodopa (L-DOPA) and/or a peripherally selective aromatic L-amino acid decarboxylase inhibitor (AADCi).

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in microparticulate form having a particular size specification with D10 equivalent circle diameter not less than 4 μm , a D50 equivalent circle diameter of 10-45 μm and a D95 equivalent circle diameter of not more than 80 μm .

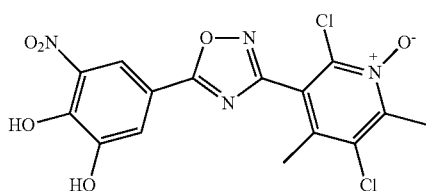
2. A compound according to claim 1, wherein the microparticulate form is prepared by ball milling or by micronisation through spiral jet mills.

3. A compound according to claim 1, wherein the compound in the microparticulate form has a particle size specification with D10 equivalent circle diameter not less than 5 μm .

4. A compound according to claim 1, wherein the compound in the microparticulate form has a particle size specification with D50 equivalent circle diameter being 15-30 μm .

5. A compound according to claim 3, wherein the compound in the microparticulate form has a particle size specification with D50 equivalent circle diameter being 15-30 μm .

6. A pharmaceutical composition in unit dosage form for oral administration comprises a compound of the formula (II)



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in microparticulate form having a particular size specification with D10 equivalent circle diameter not less than 4 μm , a D50 equivalent circle diameter of 10-45 μm and a D95 equivalent circle diameter of not more than 80 μm and a pharmaceutically acceptable carrier thereof.

7. A pharmaceutical composition according to claim 6, wherein the unit dosage form is a tablet or a capsule.

8. A method of treating Parkinson's disease in a patient suffering therefrom comprises administering to the patient an effective amount of a pharmaceutical composition according to claim 6.

9. A method according to claim 8, wherein the unit dosage form is a tablet or a capsule.

10. A compound according to claim 1, wherein the compound in the microparticulate form has a particle size specification with D10 equivalent circle diameter not less than 6 μm .

11. A compound according to claim 1, wherein the compound in the microparticulate form has a particle size specification with D50 equivalent circle diameter not less than 20-25 μm .

12. A compound according to claim 3, wherein the compound in the microparticulate form has a particle size specification with D50 equivalent circle diameter not less than 20-25 μm .

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